LECTURE 17: DIFFERENCE IN DIFFERENCES WITH HETEROGENOUS EFFECTS

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SET UP

- Data on:
  - Y_{it}, outcome, for units $i = 1, \ldots, N$, time periods $t = 1, \ldots, T$, (could be averages, say individuals within state/years)
  - $W_{it} \in \{0, 1\}$, treatment,
  - Z_i time-invariant covariates, exogenous, pre-treatment,
  - X_{it} time varying covariates, exogenous (could be more controversial).
- Standard DID setting: $W_{it} = 1$ for units $i \in \mathcal{I} \subset \{1, \ldots, N\}$ (treatment group), $t > T_0$ (post period), leading to

$$\hat{\tau}^{\mathrm{DID}} = \left(\overline{Y}^{\text{post,treatment}} - \overline{Y}^{\text{post,control}}\right) - \left(\overline{Y}^{\text{pre,treatment}} - \overline{Y}^{\text{pre,control}}\right)$$

$$= \left(\overline{Y}^{\text{post,treatment}} - \overline{Y}^{\text{pre,treatment}}\right) - \left(\overline{Y}^{\text{post,control}} - \overline{Y}^{\text{pre,control}}\right)$$

TWO-WAY-FIXED-EFFECT ESTIMATOR

- Alternative characterization for $\hat{\tau}^{\text{did}}$ as two-way-fixed-effect estimator $\hat{\tau}^{\text{twfe}}$ based on:

$$\min_{\alpha_i,\beta_t,\mu,\tau} \sum_{i=1}^{N} \sum_{t=1}^{N} (Y_{it} - \mu - \alpha_i - \beta_t - \tau W_{it})^2$$

(with normalization on α_i, *eg*, $\alpha_1 = 0$ or $\sum_i \alpha_i = 0$, and similar for β_t.)

  - TWFE estimator works for general assignment mechanism W_{it} (that does not split into treatment/control group, pre-post period).
  - Least squares estimators $\hat{\alpha}_i$ are not consistent under fixed-T large-N asymptotics, but $\hat{\tau}$ is consistent for τ.

STAGGERED ADOPTION

- Here we look at staggered adoption case, where

$$W_{it} \geq W_{it-1}$$

for all $t = 2, \ldots, T$, all $i = 1, \ldots, N$.

- Define

$$A_i \equiv \arg\min_t \mathbf{1}_{W_{it}=1}$$

the adoption date, equal to $W_i = \infty$ for never-adopters, so $W_{it} = \mathbf{1}_{t \geq A_i}$.

  - What is interpretation of $\hat{\tau}^{\text{twfe}}$ in staggered adoption setting?
  - What are alternative estimators in staggered adoption setting?

STAGGERED ADOPTION: DEFINITIONS AND NOTATION

- Now view adoption date A_i as (multi-valued) treatment.
- $Y_{it}(a)$ is the potential outcome for unit i in period t given adoption date a.
- $Y_{i,t} \equiv Y_{i,t}(A_i)$ is realized outcome for unit i in period t.
- $\overline{Y}_t(a) \equiv \frac{1}{N}\sum_{i=1}^{N} Y_{it}(a)$ is average period t potential outcome for adoption date a.
- $\overline{Y}_{t,a} \equiv \sum_{i:A_i=a} Y_{i,t} / \sum_{i:A_i=a} 1$ is average realized outcome in period t for units with adoption date a.
- $\dot{W}_{it} \equiv W_{it} - \overline{W}_{\cdot t} - \overline{W}_{i\cdot} - \overline{\overline{W}}$ doubly-demeaned treatment indicator.
- $V_W \equiv \sum_{i,t} \dot{W}_{it}^2 / (NT)$

STAGGERED ADOPTION: DEFINITIONS AND NOTATION

- Basic estimand:

$$\tau_t^{a,a'} \equiv \frac{1}{N_a} \sum_{i:A_i=a} \Big(Y_{i,t}(a) - Y_{i,t}(a') \Big)$$

- Average effect, which could reflect heterogeneity in the effects.

- Average effect
  - at time t
    - ★ could be different at other times
  - for units who adopt at time a, $a \leq t$
    - ★ could be different for other subpopulations
  - of adopting at a *versus* adopting at a', for $a \leq t < a'$
    - ★ could be different for other comparisons

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Consider adoption dates a and a', and time periods t and s, with
  - $s < a \leq t$ (so group a adopts between s and t
  - $a' \leq s$ (so group a' adopts before s) or $a' > t$ (so group a' after t),
- The basic DID estimator for these two groups (defined by the adoption dates a and a') and two time periods (defined by t and s):

$$\hat{\tau}_{t,s}^{a,a'} \equiv \left(\overline{Y}_{t,a} - \overline{Y}_{s,a}\right) - \left(\overline{Y}_{t,a'} - \overline{Y}_{s,a'}\right)$$

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Goodman-Bacon (2021) introduces a decomposition of the DID/TWFE estimator:
- first, an estimator for the average effect for units who adopt at *a versus* never:

$$\hat{\tau}^{a,\infty} \equiv \left(\frac{1}{T-(a-1)} \sum_{t=a}^{T} \overline{Y}_t^a - \frac{1}{a-1} \sum_{t=1}^{a-1} \overline{Y}_t^a \right)$$

$$-\left(\frac{1}{T-(a-1)} \sum_{t=a}^{T} \overline{Y}_t^\infty - \frac{1}{a-1} \sum_{t=1}^{a-1} \overline{Y}_t^\infty \right)$$

- This is equal to

$$\frac{1}{(T-(a-1))(a-1)} \sum_{t=a}^{T} \sum_{s=1}^{a-1} \hat{\tau}_{t,s}^{a,\infty}$$

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Second, units who adopt at a vs $a' > a$ for $a \leq t < a'$ and $1 \leq s < a$:

$$\hat{\tau}^{a,a',a<a'} \equiv \left(\frac{1}{T-(a-1)} \sum_{t=a}^{a'-1} \overline{Y}_t^a - \frac{1}{a-1} \sum_{t=1}^{a-1} \overline{Y}_t^a \right)$$

$$\left(\frac{1}{T-(a-1)} \sum_{t=a}^{a'-1} \overline{Y}_t^{a'} - \frac{1}{a-1} \sum_{t=1}^{a-1} \overline{Y}_t^{a'} \right)$$

- This is equal to

$$\frac{1}{(a'-a)(a-1)} \sum_{t=a}^{a'-1} \sum_{s=1}^{a-1} \hat{\tau}_{t,s}^{a,a'}$$

THE INTERPRETATION OF $\hat{\tau}^{t} wfe$ UNDER STAGGERED ADOPTION

- Third, units who adopt at a vs $a' < a$ for $a' \leq t < T$ and $a' \leq s < a$:

$$\hat{\tau}^{a,a',a'<a} \equiv \left(\frac{1}{T-(a-1)} \sum_{t=a}^{T} \overline{Y}_t^a - \frac{1}{a-a'} \sum_{t=a'}^{a-1} \overline{Y}_t^a \right)$$

$$\left(\frac{1}{T-(a-1)} \sum_{t=a}^{T-1} \overline{Y}_t^{a'} - \frac{1}{a-a'} \sum_{t=a'}^{a-1} \overline{Y}_t^{a'} \right)$$

- This uses already treated units as controls.
- This is equal to

$$\frac{1}{(T-(a-1)(a-a')} \sum_{t=a}^{T} \sum_{s=a'}^{a-1} \hat{\tau}_{t,s}^{a,a'}$$

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Consider for $a' \leq s < a \leq t$

$$\hat{\tau}_{t,s}^{a,a'} = \left(\overline{Y}_{t,a} - \overline{Y}_{s,a}\right) - \left(\overline{Y}_{t,a'} - \overline{Y}_{s,a'}\right)$$

- Suppose $Y_{it}(0) = 0$, so parallel trends for control outcomes are satisfied.
- Suppose $a' = 1$, $a = 2$, $s = 1$, $t = 2$, $Y_{i2}(1) = 1$ for units with $A_i = 2$, $Y_{i1}(1) = 0$ for units with $A_i = 1$, and $Y_{i1}(1) = 2$ for units with $A_i = 1$.
- Then

$$\tau^{\text{did}} = (1 - 0) - (2 - 0) = -1$$

which is negative even though all unit-level treatment effects are positive.

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Goodman-Bacon:

$$\tau^{\text{did}} = \sum_{a \neq \infty} \omega^{a\infty} \hat{\tau}^{a,\infty} + \sum_{a \neq \infty, a' > a} \omega^{aa', a<a'} \hat{\tau}^{a,a',a<a'}$$

$$+ \sum_{a \neq \infty, a' < a} \omega^{aa', a'<a} \hat{\tau}^{a,a',a'<a}$$

- Note: weights sum to one:

$$\sum_{a \neq \infty} \omega^{a\infty} + \sum_{a \neq \infty, a' > a} \omega^{aa', a<a'} + \sum_{a \neq \infty, a' < a} \omega^{aa', a'<a} = 1$$

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- Consider the weights in these three terms:

$$\omega^{a\infty} = \frac{1}{V_W}\left((N_a + N_\infty)^2 \frac{N_a}{N_a + N_\infty}\frac{N_\infty}{N_a + N_\infty}\overline{W}_a(1 - \overline{W}_a)\right)$$

$$\omega^{a,a',a<a'} = \frac{1}{V_W}\left((N_a + N_\infty)(1 - \overline{W}_{a'})^2 \frac{N_a}{N_a + N_\infty}\frac{N_\infty}{N_a + N_\infty}\frac{(\overline{W}_a - \overline{W}_{a'})(1 - \overline{W}_a)}{(1 - \overline{W}_{a'})^2}\right)$$

$$\omega^{a,a',a'<a} = \frac{1}{V_W}\left((N_a + N_\infty)\overline{W}_a)^2 \frac{N_a}{N_a + N_\infty}\frac{N_\infty}{N_a + N_\infty}\frac{(\overline{W}_a - \overline{W}_{a'})\overline{W}_{a'}}{\overline{W}_a^2}\right)$$

These weights are all nonnegative.

THE INTERPRETATION OF $\hat{\tau}^{\text{twfe}}$ UNDER STAGGERED ADOPTION

- But the third term $\hat{\tau}^{a,a',a'<a}$ in the Goodman-Bacon decomposition still creates a problem for the interpretation.
- Because the early treated group a' serves as a control, a big positive treatment effect for group a' in the period between a' and a relative to the earlier period can be misinterpreted as a trend.
- This can lead to a negative estimate for $\hat{\tau}^{a,a',a'<a}$
- Roth *et al* (2023) refer to these as "forbidden" as opposed to the other, "clean," comparisons.
- A recent literature has focused on alternative estimators.
- I am not so concerned:

"We, therefore, do not view the negative weights of some estimators as necessarily disqualifying and find the terminology "clean" and "forbidden" not doing justice to the potential benefits from such methods." (Arkangelsky and Imbens, 2023)

ALTERNATIVE ESTIMATORS: DE CHAISEMARTIN & D'HAULTFOEUILLE (2020)

- Consider the one-period ahead DID estimators:

$$\hat{\tau}_{t,t-1}^{t,a} = \left(\overline{Y}_{t,t} - \overline{Y}_{t-1,t}\right) - \left(\overline{Y}_{t,a} - \overline{Y}_{t-1,a}\right)$$

$$\hat{\tau}_{+,t} = \frac{1}{T-(a-1)} \sum_{a>t} \hat{\tau}_{t-1,t}^{t,a}$$

- Define the average of these over all time periods

$$\hat{\tau}^{\mathrm{CH}} = \sum_{t=2}^{T} \frac{N_t}{N} \hat{\tau}_{+,t}$$

where N_t is the number of adopters in period t.

- Consider for $t \geq a - \delta$ the double difference

$$\hat{\tau}^{a,\infty}_{t,a-\delta-1} = \left(\overline{Y}_{t,a} - \overline{Y}_{a-\delta-1,a}\right) - \left(\overline{Y}_{t,\infty} - \overline{Y}_{a-\delta-1,\infty}\right)$$

and

$$\hat{\tau}^{a,>t}_{t,a-\delta-1} = \left(\overline{Y}_{t,a} - \overline{Y}_{a-\delta-1,a}\right) - \frac{1}{T-t}\sum_{a'=t+1}^{T}\left(\overline{Y}_{t,a'} - \overline{Y}_{a-\delta-1,a'}\right)$$

and weighted averages of these.

ALTERNATIVE ESTIMATORS: SUN & ABRAHAM (2021)

- Consider

$$\hat{\tau}_{t,a-1}^{a,\infty} = \left(\overline{Y}_{t,a} - \overline{Y}_{a-1,a}\right) - \left(\overline{Y}_{t,\infty} - \overline{Y}_{a-1,\infty}\right)$$

and define the average

$$\hat{\tau}^{\mathrm{SA}} = \sum_{t=2}^{T} \sum_{a=2}^{T} \hat{\tau}_{t,a-1}^{a,\infty} \cdot \frac{\Pr(A_i = a | 2 \leq A_i \leq T - (t - a))}{T - 1}$$

ALTERNATIVE ESTIMATORS: CREDIBILITY

- How to choose between
  - $\hat{\tau}^{t,a}_{t,t-1}$ (CH, Chaisemartin-d'Haultfoeuille)
  - $\hat{\tau}^{a,\infty}_{t,a-\delta-1}$ (CS, Callaway-Sant'Anna)
  - $\hat{\tau}^{a,>t}_{t,a-\delta-1}$ (CS', Callaway-Sant'Anna)
  - $\hat{\tau}^{a,\infty}_{t,a-1}$ (SA, Sun-Abraham)

  and the proposed averages thereof?
- What grounds do we / should one make that choice?

ALTERNATIVE ESTIMATORS: CREDIBILITY

- Not clear! Hard to see why one is more attractive than another, or how to choose.
  - Constant additive fixed effects more plausible in short run.
    $\Longrightarrow$ CH.
  - Never-treated groups may be qualitatively different from those who are treated at some point.
    $\Longrightarrow$ CH.
  - One period ahead comparisons may miss longrun effects.
    $\Longrightarrow$ SA, CS.

ALTERNATIVE ESTIMATORS: CREDIBILITY

- Choosing between CH, CS, CS', and SA is wrong question.
- There is a large set of estimators that have nonnegative weights on $\hat{\tau}_{t,s}^{a,a'}$,
- No principled way of choosing between them.

References

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